



(globals)

Project Classes Debug

exp no-1 caesar cipher.cpp

```
61 }
62
63 int main()
64 {
65     char key[50], text[100], plain[200];
66     int i, len = 0;
67     int r1, c1, r2, c2;
68
69     printf("Enter the key: ");
70     scanf("%s", key);
71
72     createMatrix(key);
73
74     printf("\nPlayfair Matrix:\n");
75
76     for(i = 0; i < 5; i++)
77     {
78         printf("%c %c %c %c %c\n",
79             matrix[i][0], matrix[i][1], matrix[i][2],
80             matrix[i][3], matrix[i][4]);
81     }
82
83     printf("\nEnter the plaintext: ");
84     scanf("%s", text);
85
86     for(i = 0; text[i] != '\0'; i++)
87     {
88         if(text[i] == 'j')
89             text[i] = 'i';
90
91         plain[len++] = toupper(text[i]);
92     }
93
94     if(len % 2 != 0)
95         plain[len++] = 'X';
96
97     plain[len] = '\0';
98
99     printf("\nEncrypted text: ");
100
101     for(i = 0; i < len; i += 2)
102     {
103         findPosition(plain[i], &r1, &c1);
104         findPosition(plain[i + 1], &r2, &c2);
105
106         if(r1 == r2)
107         {
108             printf("%c", matrix[r1][(c1 + 1) % 5]);
109             printf("%c", matrix[r2][(c2 + 1) % 5]);
110         }
111         else if(c1 == c2)
112         {
113             printf("%c", matrix[(r1 + 1) % 5][c1]);
114             printf("%c", matrix[(r2 + 1) % 5][c2]);
115         }
116         else
117         {
118             printf("%c", matrix[r1][c2]);
119             printf("%c", matrix[r2][c1]);
120         }
121     }
122
123     return 0;
124 }
```

Compiler Resources Compile Log Debug Find Results Close

Compilation results...

Abort Compilation



C:\Users\kanis\OneDrive\Desl



Enter the key: MONARCHY

Playfair Matrix:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Enter the plaintext: INSTRUMENTS

Encrypted text: GATLMZCLRQXA

Process exited after 45.69 seconds with return value 0

Press any key to continue . . .